

Active Fault-Tolerant Control for a Class of Nonlinear Systems with Sensor Faults

Youqing Wang, Donghua Zhou, S. Joe Qin, and Hong Wang

Abstract: A general active fault-tolerant control framework is proposed for nonlinear systems with sensor faults. According to their identifiability, all sensor faults are divided into two classes: identifiable faults and non-identifiable faults. In the healthy case, the control objective is such that all outputs converge to their given set-points. A fault detection and isolation module is firstly built, which can produce an alarm when there is a fault in the system and also tell us which sensor has a fault. If the fault is identifiable, the control objective remains the same as in the healthy case; while if the fault is non-identifiable, the control objective degenerates to be such that only the healthy outputs converge to the set-points. A numerical example is given to illustrate the effectiveness and feasibility of the proposed method and encouraging results have been obtained.

Keywords: Active fault-tolerant control, identifiable fault, non-identifiable fault, nonlinear system, sensor fault.

1. INTRODUCTION

The demand for productivity leads increasingly for industrial plants to operate under challenging conditions, which consequently exposes the possibility of systems fault. This is because industrial processes typically have a large number of sensors and actuators. If a fault is not detected promptly with a proper corrective action, it will degrade the process performance and in a more serious case, even results in safety problems for the plant and personnel. In this context, a fault-tolerant controller, which is capable of maintaining the performance of the closed-loop system at an acceptable level in the presence of faults, is desirable. Because of this, fault-tolerant control using model-based analysis redundancy has received significant attention recently (e.g., [1-5]).

Since sensor faults are very common in industrial

systems, fault-tolerant control for sensor faults has to be considered. Depending on whether a fault diagnosis module is used, fault-tolerant control can be divided into two kinds: passive method and active method. For the passive method, the main task is to design a fixed controller such that the closed-loop system can maintain closed loop system stability and performance, not only when all components are operational, but also in the case of faults. In [6], Yang *et al.* proposed a reliable LQG controller for linear systems with sensor faults by solving Riccati equations. In [7], a more practical model of sensor faults was introduced and H_∞ performance of the closed-loop system was considered. In [8], Yee *et al.* designed a reliable controller for discrete-time linear systems based on the iterative LMI approach.

In an active fault-tolerant system, a new control law is redesigned using desirable properties of performance and robustness that were important in the original system, but with the reduced capability of the impaired system in mind. In order to achieve effective feedback control reconfiguration, a fault diagnosis module is generally required in an active fault-tolerant system. In [9], Trunov and Polycarpou presented a learning scheme to detect and approximate sensor faults occurring in a class of nonlinear multi-input multi-output dynamical systems. The method was robust with respect to bounded modeling uncertainties. In [10], a wavelet-based approach to the abrupt fault detection and diagnosis of sensors was described. Two methods for detecting and reconstructing sensor faults using sliding mode observers were proposed in [11]. In [12], Wang *et al.* used adaptive updating rules for

Manuscript received September 12, 2007; revised March 20, 2008; accepted April 21, 2008. Recommended by Guest Editors Marcel Staroswiecki and Bin Jiang. This work was supported by the NSFC under Grants 60721003 & 60736026, and National 863 Project of China under Grant 2006AA04Z428. S. Joe Qin acknowledges the support of the Chang Jiang Professor at Tsinghua University, Beijing.

Youqing Wang and Donghua Zhou are with the Department of Automation, Tsinghua University, Beijing 100084, China. (e-mails: yq-wang03@mails.thu.edu.cn, zdh@mail.thu.edu.cn).

S. Joe Qin is with the Viterbi School of Engineering, University of Southern California, 925 Bloom Walk, HED 211 Los Angeles, CA 90089-1211, USA (e-mail: sqin@usc.edu).

Hong Wang is with the School of Electrical and Electronic Engineering, the University of Manchester, PO Box 88, Manchester, M60 1QD, UK (e-mail: hong.wang@manchester.ac.uk).

the fault detection and diagnosis of sensor gain faults in a linear time-invariant system. Based on a bilinear model and a fault-tolerant observer, an active fault-tolerant control law was proposed for a rail traction drive with sensor intermittent disconnection faults in [13]. [14] presented a fault-tolerant control for a ship propulsion benchmark, where estimated or virtual measurements were used as feedback variables. The estimator operates on a self-adjustable design model so that its outputs can be made immune to the effects of a specific set of sensor faults. In [15], Jeong *et al.* proposed a control strategy that provides fault tolerance to the major sensor faults which may occur in an interior-permanent-magnet-motor (IPMM)-based electric vehicle propulsion drive system.

In [16], identifiability of sensor faults was studied and a novel approach to fault detection and diagnosis of a class of nonlinear systems is presented. All sensor faults are divided into two classes: the conditionally identifiable faults and the conditionally detectable faults. The above results will be extended to fault-tolerant control in this paper. The formulation of considered system is the same as that considered in [16]. Similarly, all sensor faults are divided into two classes: identifiable faults and non-identifiable faults. Firstly, a fault detection observer is proposed to detect the fault. If a fault alarm is produced, then some fault isolation observers are activated to determine which sensor has a fault. If the fault is identifiable, the control objective remains the same as that in healthy case: that is to regulate all outputs to their set-points. If the fault is non-identifiable, the control objective degenerates to regulate only the healthy outputs to their set-points. All proposed observers are designed based on LMIs, which can be solved efficiently by using toolbox in MATLAB. The feasibility and effectiveness of the proposed schemes are illustrated in a numerical example.

The remainder of the paper is organized as follows. In Section 2, the problem formulation is presented. The fault detection and isolation modules are developed in Section 3. The fault diagnosis observer is presented in Section 4. The normal controller and fault-tolerant controller are designed and analyzed in Section 5. To demonstrate the effectiveness of the proposed scheme, Section 6 gives some simulation results. Finally, conclusions are drawn in Section 7.

2. PROBLEM FORMULATION

In this paper, the nonlinear dynamical systems under study are described as follows:

$$\begin{cases} \dot{x}(t) = Ax(t) + g(x, u, t) + Bu(t) \\ y(t) = Cx(t) \end{cases} \quad (1)$$

where $x \in \mathbb{R}^n$ is the state vector; $u \in \mathbb{R}^p$ is the

input vector; $y \in \mathbb{R}^m$ is the output vector; $g(x, u, t)$ is a smooth vector field on $\mathbb{R}^n$; A, B, C are known parameter matrices of appropriate dimensions. For convenience, the controlled output is chosen as

$$z(t) = Cx(t), \quad (2)$$

which is a fictitious variable of the state vector.

Bias faults and gain faults are two kinds of sensor faults commonly existing in practice. A sensor bias fault can be described as:

$$y(t) = z(t) + f(t) \quad (3)$$

and a sensor gain fault can be described as:

$$y(t) = (1 - \Gamma(t))z(t), \quad (4)$$

where $0 \leq \Gamma(t) \leq I$ is called attenuation matrix. Obviously, equation (4) can be rewritten as

$$y(t) = z(t) - \Gamma(t)z(t) = z(t) + f(x, t). \quad (5)$$

Therefore, the above two kinds of sensor faults can be uniformly described as

$$y(t) = z(t) + f(x, t). \quad (6)$$

In this paper, both bias and gain faults are studied.

Since $y \in \mathbb{R}^m$, there are m sensor faults to be considered. In practice, it is infrequent that two or more than two sensor faults occur simultaneously. Therefore, the following assumptions are made.

Assumption 1: Only one single fault might occur at one time.

If there is a fault in sensor i , then

$$y_i(t) = z_i(t) + f_i(x, t) = C_i x(t) + f_i(x, t), \quad (7)$$

where $C_i \in \mathbb{R}^{1 \times n}$ denotes the i th row vector of matrix C .

Assumption 2: (A, C) is detectable.

Assumption 3: $g(x, u, t)$ is globally Lipschitz with respect to x , i.e.,

$$\|g(x_1, u, t) - g(x_2, u, t)\| \leq \lambda \|x_1 - x_2\|, \forall u, t, \quad (8)$$

where λ is the Lipschitz constant.

The control objective of this paper is as follows: if there is no fault, outputs $y = z$ should be regulated to achieve their given set-points y_r ; if a fault occurs and it is identifiable, the control objective is to regulate z to track set-points y_r ; if a non-identifiable fault occurs, only the unpolluted outputs are regulated to track the given set-points.

3. FAULT DETECTION AND ISOLATION

To achieve the control objective, an FDI module should be designed first to detect the fault, to isolate which sensor has fault, to judge whether the fault is identifiable.

Firstly, a fault detection observer (FDO) is introduced in the follows:

$$\dot{\bar{x}}(t) = A\bar{x}(t) + Bu + g(\bar{x}, u, t) + \bar{L}(C\bar{x} - y). \quad (9)$$

Let the error vector be defined as

$$e_x(t) \triangleq \bar{x}(t) - x(t). \quad (10)$$

From (1), (9) and (10), it can be obtained that

$$\dot{e}_x(t) = (A + \bar{L}C)e_x(t) + (g(\bar{x}, u, t) - g(x, u, t)). \quad (11)$$

The following lemma gives a sufficient condition for (11) being stable.

Lemma 1: Estimation error e_x converges asymptotically to zero if there exist matrices $R = R^T > 0$, X and positive scalar $\mu > 0$ such that the following linear matrix inequality (LMI) is satisfied:

$$\begin{bmatrix} RA + A^T R + XC + C^T X^T + \mu\lambda^2 I & R \\ R & -\mu I \end{bmatrix} < 0. \quad (12)$$

The observer gain matrix can be selected as

$$\bar{L} = R^{-1}X. \quad (13)$$

Proof: Consider the following Lyapunov function

$$V(t) = e_x^T(t) R e_x(t). \quad (14)$$

Computing $\dot{V}(t)$ along the trajectory of (11) yields

$$\begin{aligned} \dot{V} &= e_x^T \left(RA + R\bar{L}C + A^T R + C^T \bar{L}^T R \right) e_x \\ &\quad + 2e_x^T R (g(\bar{x}, u, t) - g(x, u, t)) \\ &\leq e_x^T \left(RA + R\bar{L}C + A^T R + C^T \bar{L}^T R \right) e_x \\ &\quad + \frac{1}{\mu} e_x^T R^2 e_x + \mu \|g(\bar{x}, u, t) - g(x, u, t)\|^2 \\ &\leq e_x^T \left(RA + R\bar{L}C + A^T R + C^T \bar{L}^T R \right) e_x \\ &\quad + \frac{1}{\mu} e_x^T R^2 e_x + \mu\lambda^2 \|e_x\|^2 \\ &= e_x^T \left(\begin{array}{c} RA + R\bar{L}C + A^T R \\ + C^T \bar{L}^T R + \mu\lambda^2 I + \frac{1}{\mu} R^2 \end{array} \right) e_x. \end{aligned} \quad (15)$$

Substituting (13) into (15), it can be further obtained that

$$\dot{V} \leq e_x^T (RA + XC + A^T R + C^T X^T + \mu\lambda^2 I + \frac{1}{\mu} R^2) e_x. \quad (16)$$

If the following inequality

$$RA + XC + A^T R + C^T X^T + \mu\lambda^2 I + \frac{1}{\mu} R^2 < 0 \quad (17)$$

holds, e_x converges asymptotically to zero. From the Schur complement, (17) is equivalent to (12). This completes the proof. $\square$

Similarly, the following theorem can be readily obtained.

Theorem 1: If there exist matrices $R = R^T > 0$, X and positive scalars $\mu > 0$ and $\kappa > 0$ such that the following inequality is satisfied:

$$\begin{bmatrix} RA + A^T R + XC + C^T X^T + \mu\lambda^2 I + \kappa R & R \\ R & -\mu I \end{bmatrix} < 0 \quad (18)$$

and let

$$\bar{L} = R^{-1}X \quad (19)$$

then, estimation error e_x converges exponentially to zero with rate $\kappa/2 > 0$.

Proof: Similar to the proof of Lemma 1, from (18) and (16), it can be obtained that

$$\dot{V}(t) \leq -\kappa e_x^T R e_x = -\kappa V(t). \quad (20)$$

Hence,

$$V(t) \leq e^{-\kappa t} V(0). \quad (21)$$

From (14), it can be obtain that

$$\lambda_{\min}(R) \|e_x(t)\|^2 \leq e^{-\kappa t} \lambda_{\max}(R) \|e_x(0)\|^2, \quad (22)$$

where $\lambda_{\min}(R)$ and $\lambda_{\max}(R)$ denote the minimum and maximum eigenvalues of matrix R respectively. Therefore the nom of the error vector satisfies

$$\|e_x(t)\| \leq \sqrt{\lambda_{\max}(R)/\lambda_{\min}(R)} \|e_x(0)\| e^{-\kappa t/2}. \quad (23)$$

This finished the proof.

From (9) and (1), the detection residual can be defined as

$$r(t) \triangleq \|C\bar{x}(t) - y(t)\|. \quad (24)$$

From (10) and (23), it can be seen that the following inequality holds in healthy case:

$$r(t) \leq \sqrt{\lambda_{\max}(R)/\lambda_{\min}(R)} \|C\| \|e_x(0)\| e^{-\kappa t/2}. \quad (25)$$

Note that $\|e_x(0)\|$ is unknown generally, so the threshold in (25) is unavailable. However, since it can be shown that $\|C\|\|e_x(0)\| \approx \|r(0)\|$, the fault detection can be performed using the following mechanism:

$$r(t) \begin{cases} \leq \sqrt{\lambda_{\max}(R)/\lambda_{\min}(R)} \|r(0)\| e^{-\kappa t/2}, & \text{there is no fault} \\ > \sqrt{\lambda_{\max}(R)/\lambda_{\min}(R)} \|r(0)\| e^{-\kappa t/2}, & \text{there is a fault.} \end{cases} \quad (26)$$

Notice that matrix inequality (18) is not an LMI and is therefore difficult to solve. The following algorithm is proposed here to solve this inequality:

Algorithm 1:

- 1) Solve LMI (12) and obtain $R > 0$, X and $\mu > 0$. Then R , X and μ are fixed and let $\kappa = 0$.
 - 2) Let $\kappa = \kappa + \Delta\kappa$.
 - 3) Solve (18): if solvable, go to 2); else, go to 4).
 - 4) $\kappa = \kappa - \Delta\kappa$. Exit,
- where $\Delta\kappa > 0$ is the designed search step.

Remark 1: It is worthwhile to point out that uncertainties were not considered in fault detection scheme (26). In fact, there are many literatures about robust fault detection (for example [17-19]). Since fault detection is not the emphasis of this paper, a simple fault detection threshold is chosen as

$$\varepsilon(t) \triangleq \vartheta \sqrt{\lambda_{\max}(R)/\lambda_{\min}(R)} \|r(0)\| e^{-\kappa t/2} + \eta, \quad (27)$$

where $\vartheta > 1$ and $\eta > 0$ is determined by the magnitude of uncertainties.

Because any sensor fault can affect all estimation errors simultaneously, the above observer and scheme are only used for fault detection. In the following, some observers will be designed to isolate sensor faults. For this purpose, define $\tilde{C}_i \triangleq C \setminus C_i$, i.e., it is the remaining part of matrix C with by omitting its row vector C_i . Similarly to (9) and (10), we can design m isolation observers as follows,

$$\Xi_i : \begin{cases} \dot{\tilde{x}}_i(t) = A\tilde{x}_i(t) + Bu + g(\tilde{x}_i, u, t) + \bar{L}_i(\tilde{C}_i\tilde{x}_i - \tilde{y}_i) \\ e_{xi}(t) \triangleq \tilde{x}_i(t) - x(t), \end{cases} \quad i = 1, \dots, m \quad (28)$$

where

$$\tilde{y}_i \triangleq y \setminus y_i, i = 1, \dots, m. \quad (29)$$

From Lemma 1, it is obtained that if LMI

$$\begin{bmatrix} R_i A + A^T R_i + X_i \tilde{C}_i + \tilde{C}_i^T X_i^T + \mu_i \lambda^2 I & R_i \\ R_i & -\mu_i I \end{bmatrix} < 0 \quad (30)$$

holds, with the observer gain matrices being determined by

$$\bar{L}_i = R_i^{-1} X_i, \quad (31)$$

then e_{xi} converges asymptotically to zero.

It is easy to prove that a necessary condition for (30) is $(A, \tilde{C}_i)$ being detectable.

From Theorem 1, it is obtained that if LMI

$$\begin{bmatrix} R_i A + A^T R_i + X_i \tilde{C}_i + \tilde{C}_i^T X_i^T + \mu_i \lambda^2 I + \kappa_i R_i & R_i \\ R_i & -\mu_i I \end{bmatrix} < 0 \quad (32)$$

then, estimation error e_{xi} converges exponentially to zero with rate $\kappa_i/2 > 0$. Therefore, if there is no fault, then

$$\begin{aligned} r_i(t) &\triangleq \|C_i \bar{x}_i(t) - y_i(t)\| \\ &\leq \sqrt{\lambda_{\max}(R_i)/\lambda_{\min}(R_i)} \|C_i\| \|e_{xi}(0)\| e^{-\kappa_i t/2} \end{aligned} \quad (33)$$

and

$$\begin{aligned} \tilde{r}_i(t) &\triangleq \|\tilde{C}_i \bar{x}_i(t) - \tilde{y}_i(t)\| \\ &\leq \sqrt{\lambda_{\max}(R_i)/\lambda_{\min}(R_i)} \|\tilde{C}_i\| \|e_{xi}(0)\| e^{-\kappa_i t/2}. \end{aligned} \quad (34)$$

Assume that there is a fault in sensor i and there is no fault in other sensors. Since y_i is not used in Ξ_i , so inequality (34) is still true whilst (33) does not hold. Hence, the isolation law for sensor i can be designed as

$$\text{if } \begin{cases} r_i(t) > \vartheta_i \sqrt{\lambda_{\max}(R_i)/\lambda_{\min}(R_i)} \|r_i(0)\| e^{-\kappa_i t/2} + \eta_i \\ \tilde{r}_i(t) \leq \vartheta_i \sqrt{\lambda_{\max}(R_i)/\lambda_{\min}(R_i)} \|\tilde{r}_i(0)\| e^{-\kappa_i t/2} + \eta_i, \end{cases} \quad (35)$$

then there is a fault in sensor i .

4. FAULT DIAGNOSIS

According to the fault detection and isolation results in the above section, we have obtained that whether there exist faults and which sensor has a fault. Assume that there is a fault in sensor i , an observer should be designed to estimate this fault and the real value of the polluted output. The i th output can be described as:

$$y_i(t) = C_i x(t) + f_i(x, t). \quad (36)$$

Rewriting the system description (1) by taking into account (36), we obtain

$$\begin{cases} \dot{x}(t) = Ax(t) + g(x, u, t) + Bu(t) \\ y(t) = Cx(t) + D_i f_i(x, t), \end{cases} \quad (37)$$

where it has been denoted that

$$D_i = [0 \quad \dots \quad 0 \quad 1 \quad 0 \quad \dots \quad 0]^T. \quad (38)$$

To use the observer presented in [16], matrix D_i should satisfy the following equation:

$$\text{rank} \begin{bmatrix} \mu I - A & 0 \\ C & D_i \end{bmatrix} = n + \text{rank} D_i, \quad (39)$$

$$\forall \mu \in C^+ (\text{closed right-half plane}).$$

Denote

$$\begin{cases} E = [I_n & 0_{n \times 1}] \\ M = [A & 0_{n \times 1}] \\ H_i = [C & D_i] \\ \zeta_i = \begin{pmatrix} x \\ f_i \end{pmatrix}. \end{cases} \quad (40)$$

Let

$$\begin{bmatrix} P_i & Q_i \end{bmatrix} = \left(\begin{bmatrix} E \\ H_i \end{bmatrix}^T \begin{bmatrix} E \\ H_i \end{bmatrix} \right)^{-1} \begin{bmatrix} E \\ H_i \end{bmatrix}^T, \quad (41)$$

where $P_i \in \mathbb{R}^{(n+1) \times n}$, $Q_i \in \mathbb{R}^{(n+1) \times m}$.

Construct the unknown-input observer as follows:

$$\begin{cases} \dot{z}_i = N_i z_i + L_i y + P_i B u + P_i g(\hat{x}_i, u, t), \\ \hat{\zeta}_i = z_i + Q_i y, \end{cases} \quad (42)$$

where $z_i \in \mathbb{R}^{n+1}$. Let

$$\zeta_i \triangleq \begin{pmatrix} x \\ f_i \end{pmatrix}, \hat{\zeta}_i \triangleq \begin{pmatrix} \hat{x}_i \\ \hat{f}_i \end{pmatrix}, \quad (43)$$

$$e_i = \hat{\zeta}_i - \zeta_i = \begin{pmatrix} \hat{x}_i - x \\ \hat{f}_i - f_i \end{pmatrix} \triangleq \begin{pmatrix} \tilde{x}_i \\ \tilde{f}_i \end{pmatrix}. \quad (44)$$

It is valuable to point out that $\hat{x}_i$ denotes the estimation of x produced by the i th observer.

From [16], we obtain

$$\begin{aligned} \dot{e}_i &= N_i e_i + (N_i + F_i H_i - P_i M) \zeta_i \\ &\quad + P_i (g(\hat{x}_i, u, t) - g(x, u, t)), \end{aligned} \quad (45)$$

where

$$F_i \triangleq L_i - N_i Q_i. \quad (46)$$

Let

$$N_i \triangleq P_i M - F_i H_i, \quad (47)$$

then (45) can be further expressed as

$$\dot{e}_i = (P_i M - F_i H_i) e_i + P_i (g(\hat{x}_i, u, t) - g(x, u, t)). \quad (48)$$

Lemma 2 [16]: Assume that system (1) satisfies Assumptions 2 and 3. The estimation error e_i of observer (48) converges asymptotically to zero if there exist matrices $R_i = R_i^T > 0$, X_i and positive scalars $\mu_i > 0$ such that the following LMI is satisfied:

$$\begin{aligned} \Omega_i &\triangleq \\ &\begin{bmatrix} R_i P_i M + M^T P_i^T R_i - X_i H_i - H_i^T X_i^T + \mu_i \lambda^2 I & R_i P_i \\ P_i^T R_i & -\mu_i I \end{bmatrix} \\ &< 0. \end{aligned} \quad (49)$$

where λ is the Lipschitz constant defined in (8). Let

$$F_i \triangleq R_i^{-1} X_i. \quad (50)$$

From (50), (47) and (46), we can obtain N_i and L_i .

By using (49), we can judge whether the fault can be estimated. If $\Omega_i < 0$ is solvable, then sensor i is said to be identifiable sensor, and a fault in this sensor is called identifiable fault. Otherwise, sensor i is called non-identifiable sensor and its fault is called non-identifiable fault. For each identifiable sensor, an unknown-input observer (42) is designed correspondingly. Since $y \in \mathbb{R}^m$, at most m estimation observers should be designed. In the following section, different control schemes will be designed for different classes of sensor faults.

5. CONTROLLER DESIGN

Generally speaking, m outputs could be regulated by using m inputs. Hence, the following assumption is made.

Assumption 4: It is assumed that $p = m$, and matrix CB is nonsingular.

That is to say, the considered system is a multi-input and multi-output nonlinear square system. Most industrial processes are designed to be open-loop stable; that is to say, the outputs must be bounded when the system is subjected to bounded inputs. On the other hand, due to some physical limitations, the system states are bounded by some known boundary. Hence, it is also reasonable to introduce the following assumption:

Assumption 5: In all cases, system states are bounded, i.e.,

$$\|x\| \leq \Sigma, \quad (51)$$

where $\Sigma > 0$ is a known constant.

Assume that the set-point of the output is y_r , then,

it is intuitive to choose the sliding mode as

$$s(t) \triangleq Cx(t) - y_r. \quad (52)$$

Since C is of a full row rank, there must exist a matrix $\bar{C} \in \mathbb{R}^{(n-p) \times n}$ such that matrix

$$T \triangleq \begin{bmatrix} C \\ \bar{C} \end{bmatrix} \quad (53)$$

is nonsingular. Correspondingly, the inverse matrix of T is defined as

$$T^{-1} \triangleq [T_1 \quad T_2]. \quad (54)$$

From (1), (52), (53) and (54), we obtain

$$\begin{aligned} \dot{s}(t) &= CAx(t) + Cg(x, u, t) + CBu(t) \\ &= CAT^{-1}Tx(t) + Cg(T^{-1}Tx, u, t) + CBu(t) \\ &= CA \begin{bmatrix} T_1 & T_2 \end{bmatrix} \begin{bmatrix} C \\ \bar{C} \end{bmatrix} x \\ &\quad + Cg \left(\begin{bmatrix} T_1 & T_2 \end{bmatrix} \begin{bmatrix} C \\ \bar{C} \end{bmatrix} x, u, t \right) + CBu(t) \\ &= CAT_1s(t) + CAT_1y_r + CAT_2\bar{C}x \\ &\quad + Cg(T_1s + T_1y_r + T_2\bar{C}x, u, t) + CBu(t). \end{aligned} \quad (55)$$

From the above statement, we can design the control law as the following form:

$$\begin{aligned} u &= -(CB)^{-1}CAT_1s - \alpha(CB)^{-1}s \\ &\quad - (CB)^{-1} \left[\begin{array}{l} \|CAT_1y_r\| + \|CAT_2\bar{C}\|X + \|C\| \|g(T_1s, u, t)\| \\ + \lambda \|C\| \|T_1y_r\| + \lambda \|C\| \|T_2\bar{C}\|X \end{array} \right] \text{sgn}(s), \end{aligned} \quad (56)$$

where $\alpha > 0$ and

$$\text{sgn}(s) \triangleq [\text{sgn}(s_1) \quad \text{sgn}(s_2) \quad \cdots \quad \text{sgn}(s_p)]^T. \quad (57)$$

Obviously, we have

$$s^T \text{sgn}(s) = \|s\|. \quad (58)$$

Theorem 2: Under control law (56), the output of the closed-loop system convergences to the sliding-mode surface $s = 0$ asymptotically.

Proof: Define

$$W(t) = \frac{1}{2}s^Ts \quad (59)$$

then, from (55), it can be formulated that

$$\dot{W} = s^T \dot{s} = s^T \begin{bmatrix} CAT_1s + CAT_1y_r + CAT_2\bar{C}x \\ + Cg(T_1s + T_1y_r + T_2\bar{C}x, u, t) + CBu \end{bmatrix}$$

$$\begin{aligned} &\leq s^T CAT_1s + \|s\| \|CAT_1y_r\| + \|s\| \|CAT_2\bar{C}\| \Sigma \\ &\quad + \|s\| \|C\| \|g(T_1s + T_1y_r + T_2\bar{C}x, u, t)\| + s^T CBu. \end{aligned} \quad (60)$$

From Assumption 3, the following inequality should hold true.

$$\begin{aligned} &\|g(T_1s + T_1y_r + T_2\bar{C}x, u, t)\| \\ &\leq \|g(T_1s, u, t)\| + \lambda \|T_1y_r\| + \lambda \|T_2\bar{C}\|X \end{aligned} \quad (61)$$

Combining (60) with (61), (56) and (58), it can be obtained that

$$\dot{W} \leq -\alpha s^Ts. \quad (62)$$

This completes the proof. $\square$

From Theorem 2 and (52), we obtain that the outputs convergence to their set-points asymptotically.

In designing sliding mode control law (56), it is assumed that the control can switch infinitely fast. This is infeasible due to speed limitations on actuators. The non-ideal switching can result in chattering phenomenon. To eliminate this undesirable chattering, it is practical to replace the sign function by a saturation function, $\text{sat}_\mu(s)$, which is defined by

$$\begin{aligned} \text{sat}_\mu(s) &\triangleq [\text{sat}_\mu(s_1) \quad \text{sat}_\mu(s_2) \quad \cdots \quad \text{sat}_\mu(s_p)]^T, \\ \text{sat}_\mu(s_i) &\triangleq \begin{cases} s_i/\mu, & \text{if } |s_i/\mu| < 1 \\ \text{sgn}(s_i/\mu), & \text{if } |s_i/\mu| \geq 1, \end{cases} \quad i=1, \dots, p. \end{aligned} \quad (63)$$

Theorem 3: Under the following control law

$$\begin{aligned} u &= -(CB)^{-1}CAT_1s - \alpha(CB)^{-1}s \\ &\quad - (CB)^{-1} \left[\begin{array}{l} \|CAT_1y_r\| + \|CAT_2\bar{C}\| \Sigma + \lambda \|C\| \|T_1y_r\| \\ + \lambda \|C\| \|T_2\bar{C}\| \Sigma + \|C\| \|g(T_1s, u, t)\| \end{array} \right] \text{sat}_\mu(s) \end{aligned} \quad (64)$$

the closed-loop system convergences to boundary layer $|s| \leq \mu$ ultimately. In other words, sliding mode $s(t)$ is ultimately uniformly bounded.

Proof: From the proof of Theorem 2 and (63), we obtain that $\dot{W} \leq -\alpha s^Ts$ if $|s/\mu| \geq 1$. Hence, $s(t)$ convergences to boundary layer $|s| \leq \mu$ ultimately.

If there is a fault in identifiable sensor i ($i=1, \dots, p$), then, observer (42) is used to estimate the states and fault. It is reasonable to design the fault-tolerant control law as the following form:

$$\begin{aligned} u &= -(CB)^{-1}CAT_1\hat{s}_i - \alpha(CB)^{-1}\hat{s}_i \\ &\quad - (CB)^{-1} \left[\begin{array}{l} \|CAT_1y_r\| + \|CAT_2\bar{C}\| \Sigma + \lambda \|C\| \|T_1y_r\| \\ + \|C\| \|g(T_1\hat{s}_i, u, t)\| + \lambda \|C\| \|T_2\bar{C}\| \Sigma \end{array} \right] \text{sat}_\mu(\hat{s}_i), \end{aligned} \quad (65)$$

where

$$\hat{s}_i \triangleq [s_1 \cdots s_{i-1} \quad C_i \hat{x}_i - y_r^i \quad s_{i+1} \cdots s_p]^T. \quad (66)$$

For this fault tolerant control law, the following result can be obtained

Theorem 4: Under control law (65), the closed-loop system convergences to boundary layer $|s| \leq \mu$ ultimately.

Proof: From (59), (60), (61), (63), and (65), we obtain that if $|s| \geq \mu$, then

$$\begin{aligned} \dot{W} \leq & -\alpha s^T s + s^T CAT_1 \tilde{s}_i + \alpha s^T \tilde{s}_i \\ & + s^T \|C\| [\|g(T_1 s, u, t)\| sat_\mu(s) \\ & \quad - \|g(T_1 \hat{s}_i, u, t)\| sat_\mu(\hat{s}_i)] \\ & + s^T [\|CAT_1 y_r\| + \|CAT_2 \bar{C}\| \Sigma \\ & \quad + \lambda \|C\| \|T_1 y_r\| + \lambda \|C\| \|T_2 \bar{C}\| \Sigma] \\ & \times [sat_\mu(s) - sat_\mu(\hat{s}_i)], \end{aligned} \quad (67)$$

where $\tilde{s}_i \triangleq s - \hat{s}_i$. From Lemma 2 and (66), it can shown that

$$\lim_{t \rightarrow \infty} \|\tilde{s}_i(t)\| = 0. \quad (68)$$

From Assumption 3 and (63), we obtain that $\|g(T_1 s, u, t)\| sat_\mu(s) - \|g(T_1 \hat{s}_i, u, t)\| sat_\mu(\hat{s}_i)$ and $sat_\mu(s) - sat_\mu(\hat{s}_i)$ convergence to zeros as $\|\tilde{s}_i(t)\|$ convergences to zero. On the other hand, from Assumption 5, we know that s is bounded. Therefore, there is a $T > 0$ such that if $t \geq T$ and $|s| \geq \mu$

$$\begin{aligned} & \left\| s^T CAT_1 \tilde{s}_i + \alpha s^T \tilde{s}_i + s^T \|C\| [\|g(T_1 s, u, t)\| sat_\mu(s) \right. \\ & \quad \left. - \|g(T_1 \hat{s}_i, u, t)\| sat_\mu(\hat{s}_i)] + s^T [\|CAT_1 y_r\| + \|CAT_2 \bar{C}\| \Sigma \right. \\ & \quad \left. + \lambda \|C\| \|T_1 y_r\| + \lambda \|C\| \|T_2 \bar{C}\| \Sigma] \times [sat_\mu(s) - sat_\mu(\hat{s}_i)] \right\| \\ & \leq \frac{1}{2} \alpha \mu^2 \leq \frac{1}{2} \alpha s^T s. \end{aligned} \quad (69)$$

Combining (67) and (69), it can be shown that the following inequality

$$\dot{W} \leq -\frac{1}{2} \alpha s^T s < 0 \quad (70)$$

holds, if $t \geq T$ and $|s| \geq \mu$. Hence, W will decrease, and then $s(t)$ will decrease until it convergences to boundary layer $|s| \leq \mu$ ultimately. This finishes the proof.

If there is a fault in non-identifiable sensor i ($i=1, \dots, p$), then we cannot design an estimate observer. Hence, the control objective should be released. For this purposed, define

$$\tilde{C}_i \triangleq \begin{bmatrix} C_1 \\ \vdots \\ C_{i-1} \\ C_{i+1} \\ \vdots \\ C_p \end{bmatrix}, \quad \tilde{y}_{ri} \triangleq \begin{bmatrix} y_r^1 \\ \vdots \\ y_r^{i-1} \\ y_r^{i+1} \\ \vdots \\ y_r^p \end{bmatrix}, \quad s_i \triangleq \tilde{C}_i x - \tilde{y}_{ri}. \quad (71)$$

From Assumption 4, it can be seen that there exist $j_i \in \{1, 2, \dots, p\}$, such that $\tilde{C}_i \tilde{B}_{j_i}$ is nonsingular, where

$$\tilde{B}_{j_i} \triangleq [B_1 \quad \cdots \quad B_{j_i-1} \quad B_{j_i+1} \quad \cdots \quad B_p]. \quad (72)$$

Notice that

$$Bu = \tilde{B}_{j_i} \tilde{u}_{j_i} + B_{j_i} u_{j_i},$$

where

$$\tilde{u}_{j_i} \triangleq [u_1 \quad \cdots \quad u_{j_i-1} \quad u_{j_i+1} \quad \cdots \quad u_p]^T. \quad (73)$$

Assume that the non-identifiable fault is isolated at time $T_{isolate}$, then it is reasonable to choose the j_i th control law as

$$u_{j_i} \equiv u_{j_i}(T_{isolate}). \quad (74)$$

Then, similarly to (64), we can design the fault-tolerant control law as

$$\begin{aligned} \tilde{u}_{j_i} = & -(\tilde{C}_i \tilde{B}_{j_i})^{-1} \tilde{C}_i AT_{1i} s_i - \alpha (\tilde{C}_i \tilde{B}_{j_i})^{-1} s_i \\ & - (\tilde{C}_i \tilde{B}_{j_i})^{-1} sat_\mu(s_i) \\ & \times \left[\|\tilde{C}_i AT_{1i} y_r\| + \|\tilde{C}_i AT_{2i} \bar{C}_i\| \Sigma + \|\tilde{C}_i\| \|g(T_{1i} s_i, u, t)\| \right. \\ & \quad \left. + \lambda \|\tilde{C}_i\| \|T_{1i} y_r\| + \lambda \|\tilde{C}_i\| \|T_{2i} \bar{C}_i\| \Sigma + \|\tilde{C}_i B_{j_i} u_{j_i}\| \right], \end{aligned} \quad (75)$$

where $\begin{bmatrix} \tilde{C}_i \\ \bar{C}_i \end{bmatrix}$ is nonsingular and

$$\begin{bmatrix} T_{1i} & T_{2i} \end{bmatrix} \begin{bmatrix} \tilde{C}_i \\ \bar{C}_i \end{bmatrix} = I. \quad (76)$$

Theorem 5: Under control law (75), the closed-loop system convergences to boundary layer $|s_i| \leq \mu$ ultimately.

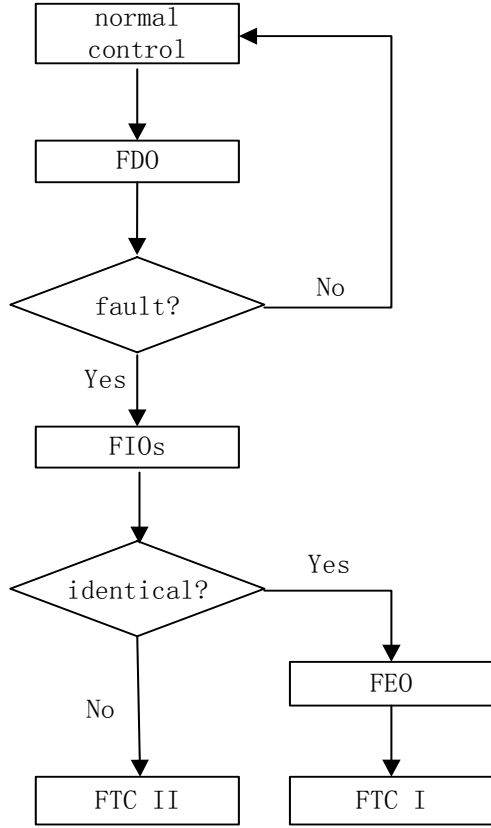


Fig. 1. Block diagram of the proposed scheme.

Proof: The proof is similar to that of Theorem 3, and is omitted here.

In this section, two kinds of fault-tolerant control laws were designed for identifiable and non-identifiable faults respectively. For convenience, control law (65) is named FTC I, whilst control law (74) and (75) is named FTC II. To summarize the controllers, the block diagram of the proposed scheme is shown in Fig. 1.

6. SIMULATION STUDY

In this section, the following example is studied

$$A = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix}, B = I_3, C = I_3, g = 0.8 \times \begin{bmatrix} \sin x_3 \\ \cos x_2 \\ \sin x_1 \end{bmatrix}.$$

Solving LMI (12), it can be obtained that

$$\bar{L} = \begin{bmatrix} -0.7150 & 0.8420 & -35.3886 \\ -1.8420 & -1.7150 & -79.0740 \\ 35.3886 & 78.0740 & -2.2150 \end{bmatrix},$$

$$R = \begin{bmatrix} 10.1542 & 0 & 0 \\ 0 & 10.1542 & 0 \\ 0 & 0 & 10.1542 \end{bmatrix}.$$

Let $\Delta\kappa = 0.01$ and use Algorithm 1, then we obtain that $\kappa = 2.66$. The detection threshold is chosen as in (27), where $\vartheta = 2$ and $\eta = 0.5$.

Solving LMI (30) and using Algorithm 1, we obtain that

$$\bar{L}_1 = \begin{bmatrix} -1.0000 & 0 \\ -1.6969 & -82.7877 \\ 81.7877 & -2.1969 \end{bmatrix},$$

$$\bar{L}_2 = \begin{bmatrix} -1.1504 & -52.3676 \\ -1.4511 & -11.9846 \\ 48.3983 & -2.1261 \end{bmatrix},$$

$$\bar{L}_3 = \begin{bmatrix} -0.0440 & -106.1783 \\ 145.4842 & -2.6692 \\ 84.7751 & -2.6742 \end{bmatrix},$$

$$R_1 = \begin{bmatrix} 12.9899 & 0 & 0 \\ 0 & 10.0428 & 0 \\ 0 & 0 & 10.0428 \end{bmatrix}, \kappa_1 = 1.71,$$

$$R_2 = \begin{bmatrix} 9.0098 & -3.2557 & 0 \\ -3.2557 & 15.5213 & 0 \\ 0 & 0 & 9.0098 \end{bmatrix}, \kappa_2 = 0.39,$$

$$R_3 = \begin{bmatrix} 0.4186 & 0 & 0 \\ 0 & 0.4840 & -0.3112 \\ 0 & -0.3112 & 0.5340 \end{bmatrix}, \kappa_3 = 0.05.$$

Solving LMI (49) for three sensors respectively, it is obtained that

$$N_1 = \begin{bmatrix} -1.4430 & -1.7896 & 0.1280 & 0.5570 \\ 2.6844 & -1.3290 & 0.2304 & 2.6844 \\ -0.1921 & -0.2304 & -1.3290 & -0.1921 \\ 0.5570 & -1.7896 & 0.1280 & -1.4430 \end{bmatrix},$$

$$L_1 = \begin{bmatrix} 0 & 1.8948 & -0.0640 \\ 0 & 0.1645 & 0.3848 \\ 0 & 0.1152 & 0.4145 \\ 0 & -0.1052 & -0.0640 \end{bmatrix},$$

$$P_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.5 \\ -1 & 0 & 0 \end{bmatrix}, Q_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.5 \\ 1 & 0 & 0 \end{bmatrix},$$

and

$$N_2 = \begin{bmatrix} -3.0817 & -1.8939 & 0.0661 & -2.3939 \\ -1.9868 & -2.8766 & 2.7964 & -1.8766 \\ -0.0514 & -3.9756 & -1.2841 & -3.9756 \\ 4.2809 & 2.0680 & 2.6749 & 1.0680 \end{bmatrix},$$

$$L_2 = \begin{bmatrix} 0.5408 & 0 & -0.0330 \\ 0.9934 & 0 & -0.3982 \\ 0.0257 & 0 & 0.3921 \\ -2.1405 & 0 & -2.3374 \end{bmatrix},$$

$$P_2 = \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0.5 \\ 0 & -1 & 0 \end{bmatrix}, \quad Q_2 = \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0.5 \\ 0 & 1 & 0 \end{bmatrix},$$

however $\Omega_3 < 0$ is unsolvable, that is to say, sensor 3 is non-identifiable sensor.

Then, the normal control law, FTC I and FTC II can be designed as (64), (65) and (74), (75) respectively, where $\alpha = 0.5$, $\Sigma = 50$ and $\mu = 0.5$. To demonstrate the effectiveness of the proposed method sufficiently, three kinds of sensor faults are studied in the following.

Case 1: Bias Fault in Sensor 1.

In this case, the measured outputs are assumed to be

$$y(t) = \begin{cases} z(t), & 0 \leq t < 5 \\ z(t) + \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}, & t \geq 5. \end{cases}$$

The output responses under FTC and normal control are shown in Fig. 2, which indicates that the proposed method can tolerate this fault excellently while the normal one cannot. The fault detection result is shown in Fig. 3(a), and it is found that the fault can be detected immediately after 5 second. The

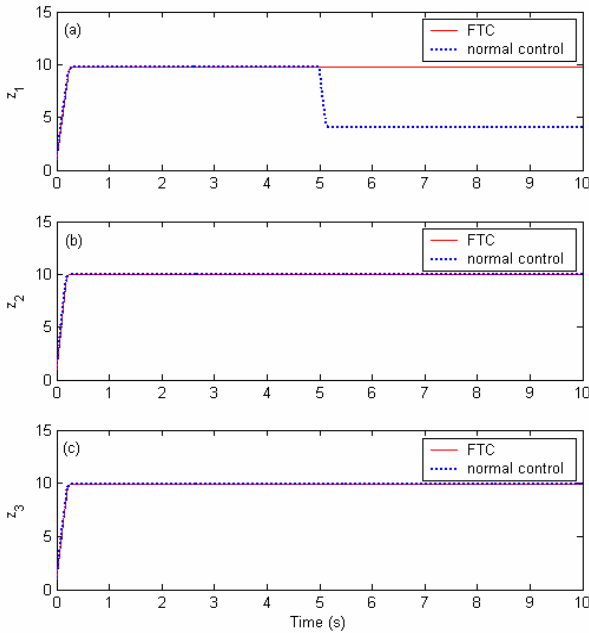


Fig. 2. Comparison of FTC and normal control with fault in sensor 1.

estimation results of isolation observer 1 is shown in Figs. 3(b) and (c). From (35), we can judge that the fault is in sensor 1, so the corresponding control law can be activated then.

Case 2: Gain Fault in Sensor 2.

The measured outputs are

$$y(t) = \begin{cases} z(t), & 0 \leq t < 5 \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.8 & 0 \\ 0 & 0 & 1 \end{bmatrix} z(t), & t \geq 5. \end{cases}$$

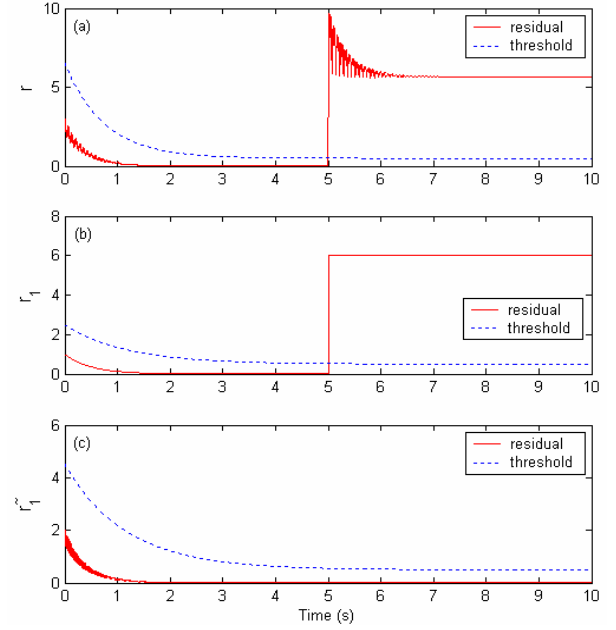


Fig. 3. Fault detection and isolation for fault in sensor 1.

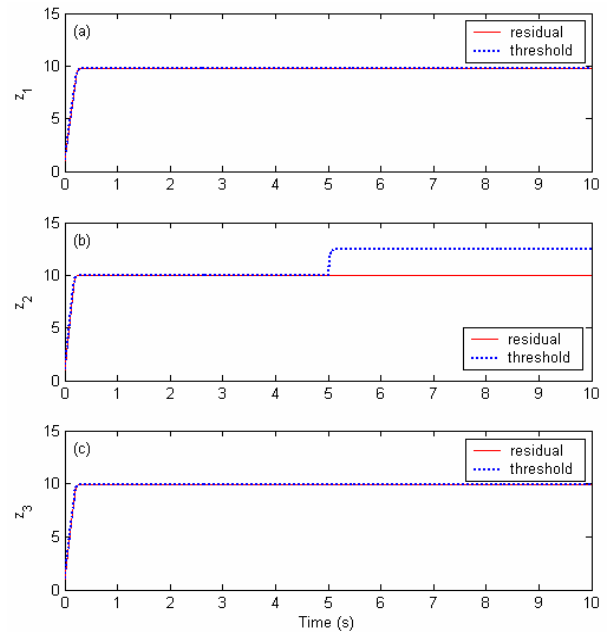


Fig. 4. Comparison of FTC and normal control with fault in sensor 2.

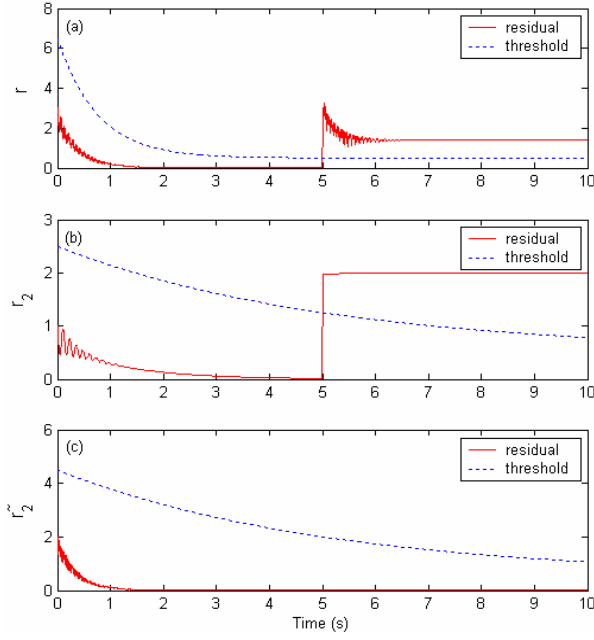


Fig. 5. Fault detection and isolation for fault in sensor 2.

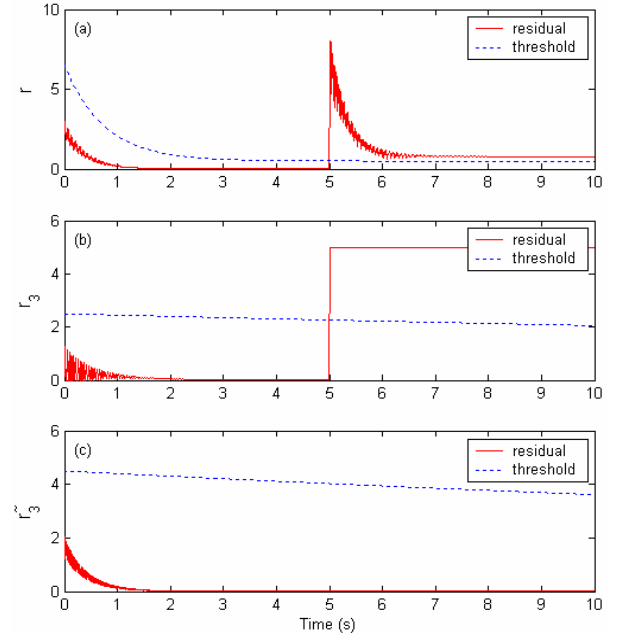


Fig. 7. Fault detection and isolation for fault in sensor 3.

7. CONCLUSIONS

By designing one detection observer for all outputs and presenting an isolation observer and an estimation observer for each output, an active fault-tolerant control framework has been proposed for a class of nonlinear systems with sensor faults. If an identifiable fault occurs, the closed-loop system under fault-tolerant control has a similar control performance compared with that in the healthy case. If a non-identifiable fault occurs, the closed-loop control performance has a little degradation. A simulated example is included to show the effectiveness of the proposed algorithms and encouraging results have been obtained.

REFERENCES

- [1] D. H. Zhou and P. M. Frank, "Fault diagnostics and fault tolerant control," *IEEE Trans. on Aerospace and Electronic Systems*, vol. 34, no. 2, pp. 420-427, Apr. 1998.
- [2] S. C. Patwardhan, S. Manuja, S. Narasimhan, and S. L. Shah, "From data to diagnosis and control using generalized orthonormal basis filters, Part II: Model predictive and fault tolerant control," *Journal of Process Control*, vol. 16, no. 2, pp. 157-175, Feb. 2006.
- [3] X. Zhang, T. Parisini, and M. M. Polycarpou, "Adaptive fault-tolerant control of nonlinear uncertain systems: An information-based diagnostic approach," *IEEE Trans. on Automatic Control*, vol. 49, no. 8, pp. 1259-1274, Aug. 2004.
- [4] Y. Q. Wang, D. H. Zhou, and F. R. Gao, "Robust

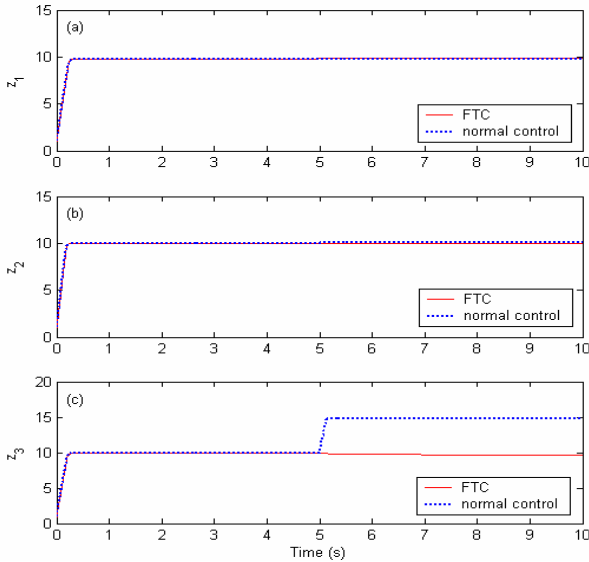


Fig. 6. Comparison of FTC and normal control with fault in sensor 3.

The output responses are shown in Fig. 4. The fault detection and isolation results are given in Fig. 5.

Case 3: Bias Fault in Sensor 3.

In this case, the measured outputs are

$$y(t) = \begin{cases} z(t), & 0 \leq t < 5 \\ z(t) + [0 \ 0 \ -5]^T, & t \geq 5. \end{cases}$$

From Fig. 6(c), we can find that the control performance of z_3 descends evidently under normal control law but slightly under FTC II. The fault detection and isolation results are shown in Fig. 7.

- fault-tolerant control of a class of non-minimum phase nonlinear processes," *Journal of Process Control*, vol. 17, no. 6, pp. 523-537, Jul. 2007.
- [5] Y. Q. Wang, J. Shi, D. H. Zhou, and F. R. Gao, "Iterative learning fault-tolerant control for batch processes," *Industrial & Engineering Chemistry Research*, vol. 45, no. 26, pp. 9050-9060, Dec. 2006.
- [6] G.-H. Yang, J. L. Wang, and Y. C. Soh, "Reliable LQG control with sensor failures," *IEE Proc.-Control Theory Appl.*, vol. 147, no. 4, pp. 433-439, Jul. 2000.
- [7] G.-H. Yang, J. L. Wang, and Y. C. Soh, "Reliable H_∞ controller design for linear systems," *Automatica*, vol. 37, no. 5, pp. 717-725, May 2001.
- [8] J.-S. Yee, G.-H. Yang, and J. L. Wang, "Reliable output-feedback controller design for discrete-time linear systems: An iterative LMI approach," *Proc. of the American Control Conference*, Arlington, VA, pp. 1035-1040, June 2001.
- [9] A. B. Trunov and M. M. Polycarpou, "Automated fault diagnosis in nonlinear multivariable systems using a learning methodology," *IEEE Trans. on Neural Network*, vol. 11, no. 1, pp. 91-101, Jan. 2000.
- [10] J. Q. Zhang and Y. Yan, "A wavelet-based approach to abrupt fault detection and diagnosis of sensors," *IEEE Trans. on Instrumentation and Measurement*, vol. 50, no. 5, pp. 1389-1396, Oct. 2001.
- [11] C. P. Tan and C. Edwards, "Sliding mode observers for detection and reconstruction of sensor faults," *Automatica*, vol. 38, no. 10, pp. 1815-1821, Oct. 2002.
- [12] H. Wang, Z. J. Huang, and S. Daley, "On the use of adaptive updating rules for actuator and sensor fault diagnosis," *Automatica*, vol. 33, no. 2, pp. 217-225, Feb. 1997.
- [13] S. M. Bennett, R. J. Patton, and S. Daley, "Sensor fault-tolerant control of a rail traction drive," *Control Engineering Practice*, vol. 7, no. 2, pp. 217-225, Feb. 1999.
- [14] N. E. Wu, S. Thavamani, Y. Zhang, and M. Blanke, "Sensor fault masking of a ship propulsion system," *Control Engineering Practice*, vol. 14, no. 11, pp. 1337-1345, Nov. 2006.
- [15] Y.-S. Jeong, S.-K. Sul, S. E. Schulz, and N. R. Patel, "Fault detection and fault-tolerant control of interior permanent-magnet motor drive system for electric vehicle," *IEEE Trans. on Industry Applications*, vol. 41, no. 1, pp. 46-51, Jan-Feb 2005.
- [16] Y. Q. Wang and D. H. Zhou, "Sensor gain fault diagnosis for a class of nonlinear systems," *European Journal of Control*, vol. 12, no. 5, pp. 523-535, Sep./Oct. 2006.
- [17] B. Jiang, M. Staroswiecki, and V. Cocquempot, "Fault estimation in nonlinear uncertain systems using robust/sliding-mode observers," *IEE Proc.-Control Theory Appl.*, vol. 151, no. 1, pp. 29-37, Jan. 2004.
- [18] X.-G. Yan and C. Edwards, "Sensor fault detection and isolation for nonlinear systems based on a sliding mode observer," *Int. J. Adapt. Control Signal Process.*, vol. 21, no. 8-9, pp. 657-673, Oct./Nov. 2007.
- [19] X.-G. Yan and C. Edwards, "Nonlinear robust fault reconstruction and estimation using a sliding mode observer," *Automatica*, vol. 43, no. 9, pp. 1605-1614, Sep. 2007.



Youqing Wang received the B.S. degree from Shandong University, China, in 2003, and the M.S. and Ph.D. degrees from the Department of Automation, Tsinghua University, in 2005 and 2008, respectively. From Feb. 2006 to Aug. 2007, he was a Research Assistant in the Dept. of Chemical Engineering, Hong Kong University of

Science and Technology, Hong Kong. Now, he works in Dept. of Chemical Engineering, University of California, Santa Barbara, US, as a Postdoctoral Fellow. Dr. Wang has published about 20 papers in international journals and conferences, including *Journal of Process Control*, *Chemical Engineering Science*, *Industrial & Engineering Chemistry Research*, *European Journal of Control* and so on. He serves as a referee for *International Journal of Control*, *Journal of Process Control*. His research interests include biomedical control, fault-tolerant control, model predictive control, batch process control, iterative learning control and fault diagnosis.



Donghua Zhou received the B.Eng., M.Sci. and Ph.D. degrees all from the Department of Electrical Engineering, Shanghai Jiaotong University, in 1985, 1988 and 1990, respectively. He was an Alexander von Humboldt Research Fellow (1995-1996) in the University of Duisburg, Germany, and a Visiting Professor, Yale University (July 2001-

Jan. 2002). He is currently a Professor and the Associate Chair in the Department of Automation, Tsinghua University. He has published over 160 journal papers and three monographs in the areas of process identification, diagnosis and control. Prof. Zhou serves the profession in many capacities such as IFAC Technical Committee of Fault Diagnosis and Safety, Deputy General Secretary of Chinese Association of Automation (CAA), Council member of CAA. He was also the NOC Chair of the 6th IFAC Symposium on SAFEPROCESS, 2006.



S. Joe Qin is the Fluor Professor of Process Engineering at the Viterbi School of Engineering at the University of Southern California and the Chang Jiang (Guest Chair) Professor affiliated with Tsinghua University appointed by the Ministry of Education of China. He obtained the B.S. and M.S. degrees in Automatic

Control from Tsinghua University in Beijing, China, in 1984 and 1987, respectively, and the Ph.D. degree in Chemical Engineering from University of Maryland in 1992. Prior to joining the Faculty at the University of Southern California he was a Professor at the University of Texas at Austin from 1995 to 2007 and a Principal Engineer at Fisher-Rosemount from 1992 to 1995. Dr. Qin's research interests include system identification, process monitoring and fault diagnosis, model predictive control, run-to-run control, semiconductor process control, and control performance monitoring. He is a Recipient of the National Science Foundation CAREER Award, DuPont Young Professor Award, Halliburton/Brown & Root Young Faculty Excellence Award, NSF-China Outstanding Young Investigator Award, and an IFAC Best Paper Prize for the model predictive control survey paper published in *Control Engineering Practice*. He is currently an Associate Editor for *Journal of Process Control* and a Member of the Editorial Board for *Journal of Chemometrics*. He served as an Editor for *Control Engineering Practice* and an Associate Editor for *IEEE Transactions on Control Systems Technology*.



Hong Wang received the B.S. degree from Huainan University of Mining Engineering, Huainan, P. R. China, in 1982, and the M.S. and Ph.D degrees from Huazhong University of Science and Technology, Huazhong, R.O.C., in 1984, 1987, respectively. From 1988 to September 1992, he was a Research Fellow at Salford, Brunel, and

Southampton Universities, U.K. He then joined University of Manchester Institute of Science and Technology (UMIST), Manchester, U.K., and has been a Professor in process control since 2002 at the University of Manchester, UK. Prof. Wang was an Associate Editor of the *IEEE Transactions on Automatic Control* and now serves as an Editorial Board Member for 6 other international Journals. Prof. Wang's research interests are stochastic distribution control, fault detection and diagnosis, nonlinear control, and data based modeling for complex systems. He has published 200 papers and 3 books in these areas.